

PARTS LIST

Semiconductors:

IC1	- ATmega328 microcontroller
IC2	- 7805, 5V regulator
IC3, IC4	- PC817 optocoupler
IC5	- TD62783 grid display driver
IC6	- TLC5916 segment display driver

LED1-LED4	- 5mm light-emitting diode
D1, D2	- 1N4007 rectifier diode
D3	- 1N4148 signal diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 680-ohm
R2, R10	- 10-kilo-ohm
R3, R4	- 1.5-kilo-ohm
R5, R6	- 1-kilo-ohm
R7	- 4.7-kilo-ohm
R8	- 330-ohm

Capacitors:

C1, C2, C5	-
C6, C10	- 0.1 μ F ceramic
C3, C4	- 22pF ceramic
C7, C8	- 1 μ F, 63V electrolytic

Miscellaneous:

CON1	- 8-pin dual-line berg strip male connector
CON2	- 4-pin SIL connector
CON3	- 16-pin berg strip male connector
CON4	- 16-pin DIL berg strip female connector
X ₅₅₅	- 16MHz crystal
DIS1-DIS3	- 44.5mm dual-colour common-anode seven-segment display

The steps to connect your Arduino board to the computer and upload the code in the MCU follow:

1. Get a USB cable. Arduino is a simple board to start working with electronics and microcontroller programming. You need a standard USB cable (A plug to B plug)—like the one you connect to a USB printer—to connect the board to the PC.

2. Download the Arduino environment. To program the Arduino board, you need the Arduino environment. Download the latest version from the Internet. When the download completes, there would be a few files and sub-folders inside a folder named Arduino. Unzip the downloaded files retaining the folder structure. Note that you should have Java JDK pre-installed in your system.

3. Locate the USB drivers. If you are using a USB-based Arduino, you need to install drivers for the FTDI



Fig. 7: Author's prototype of traffic light count-down timer with dual-colour display

chip on the board. These can be found in the drivers/FTDI USB drivers directory of the Arduino distribution. The latest version of the drivers can be found on the FTDI website.

4. Connect the board. When you connect the board to the PC using the USB cable, 'Add New Hardware' wizard pops up on the screen. Select 'Install from a list or specified location (Advanced)' option and click 'Next.' Browse to the location of USB drivers in your PC. The 'New Hardware' wizard pops up.

Repeat step 4 to install 'USB Serial Port.' Open the Windows Device Manager from 'Control Panel.' If you find 'USB Serial Port' listed under its 'Ports' section, the Arduino board is installed in your PC. After successful installation, the green power LED (PWR) on the Arduino/Freedomino board should glow.

5. Run the Arduino environment. Open the Arduino folder and double-click the Arduino application.

6. Upload a program. Open the LED blink sample file as follows:

File→Sketchbook→Examples→Digital→Blink

Select the serial port of the Arduino board from the Tools→Serial Port menu. On Windows, this should be COM1 or COM2 for a serial board, or

ATmega328 MCU.

Click 'Upload' button in the Arduino environment. After a few seconds, you will see the RX and TX LEDs on the board flashing. If the upload is successful, the message 'Done uploading' appears in the status bar.

7. Look for the blinking LED. A few seconds after the upload, the LED on pin 13 of the MCU on the board should start blinking. If it does, congratulations! You've got Arduino up-and-running.

8. Upload the traffic light dual colour timer program. Run the Arduino IDE, click File→Open (browse the location where you have saved the source code). Compile the code from Sketch→Very/Compile option. Connect Freedomino board with ATmega328 MCU and upload the program. Remove the programmed MCU from the Freedomino board and put it in the PCB of the dual-colour timer. Perform manual testing of the timer as explained earlier.

EFY note. The source code of this article has been included in this month's EFY-CD and is also available on the www.efymag.com website.

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